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Simultanuous DNA and RNA extraction from various tissues of the two oyster species, Crassostrea gigas and Ostrea edulis, for metabarcoding

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Abstract

Metabarcoding is a widely used approach to assess the microbiome of any organism of interest. Whereas DNA is widely used for this approach, the inclusion of RNA (cDNA) can add valuable information about the live fraction of an organisms' microbiome.
The very first and crucial step hereby is the extraction of nucleic acids. Often, the amount of biological material is limited and a high number of samples need to be processed at the same time. It is therefore desirable to have an extraction method available that allows the simultaneous and high-throughput extraction of DNA and RNA.

Guidelines

The AllPrep® 96 DNA/RNA kit, designed to extract four 96-well plates, provides enough plastic consumables (S-Blocks, AirPore Tape Sheets) for one extraction of two 96-well plates. Although the supplier states that those plastic consumables can be re-used, it is recommended to purchase additional S-Blocks and AirPore Tape Sheets in order to avoid cross-contamination.

An extraction of two 96-well plates usually takes 4-5 hours.

Materials

- AllPrep 96 DNA/RNA kit (ref 80311-Qiagen)
- Centrifuge Eppendorf 5804R with rotor A-2DWP (max 2250g, max 3700 rpm)
- Multichannel micropipette EPPENDORF Xplorer PLUS VARIABLE multicanaux 50-1200
- Multichannel micropipette EPPENDORF Xplorer PLUS VARIABLE multicanaux 50-1200
- Sterile Filter tips 1000 µl (ref S1122-1730C Starlab)
- Sterile Filter tips 100 µl (ref S1123-1740 Starlab)
- Ethanol absolute (ref E/0650DF/15-Fisher Scientific)
- Ethanol 70% in DNase/RNase free ultrapure water (ref 10977-035- Invitrogen Fisher Scientific)
- AirPore Tape Sheets Qiagen (need 22 sheets per extraction of 2 96 well plates, ref 19571-Qiagen)
- Qiagen S-Blocks (need 4 blocks per extraction of 2 96 well plates, ref 19585-Qiagen)
- DNA away (ref 7010 – Molecular BioProducts)
- Stainless steel beads 5 mm (ref 69989-Qiagen)
- RNase away (ref 7000 – Molecular BioProducts)
- Collection microtubes (ref 19560-Qiagen)
- Collection microtubes caps (ref 19566-Qiagen)
- Sterile reagent reservoir for multichannel pipette (ref E2310-1010-Starlab)
- Beta-mercaptoethanol (ref M3148 – Sigma-Aldrich)
- TissueLyser II (ref 85300-Qiagen) with adapter set 2-96 (ref 69984 Qiagen)

Troubleshooting

Safety warnings

Work under fume hood

Before start

- Autoclave stainless steel beads.
- Collection microtubes and collection microtube caps are placed under UV radiation for 30 min in order to minimize contamination.
- The day before the extraction, prepare buffers and solutions needed for the DNA/RNA extraction 70% ethanol, buffer RLT+beta-mercaptoethanol (βME), buffer RPE, buffer AW1
- prewarm buffer EB at 70°C

Sample preparation prior to nucleic acid extraction

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Preferentially, the experimental plan for the nucleic acid extraction is already established when samples are collected which ensures best possible sample storage and speeds up subsequent sample processing. However, if the experimental plan is not yet available at the time of collection, the samples will have to be prepared for subsequent nucleic acid extraction:
Tools needed for sample preparation (e.g. pipette, foreceps) are wiped with RNase away. Under sterile conditions (laminar flow), oyster tissues (digestive gland (DG), gills (G)) are transferred into microcollection tubes containing one 5 mm stainless steel bead according to the pre-defined plate layout. The tubes are closed with strip caps and stored at -80°C until further processing (note that each cap should be pierced with a fine needle in order to avoid pressure build-up). Tissues need to stay frozen at all times during sample preparation (put collection microtubes in the corresponding microtube racks and place in liquid nitrogen). Samples of oyster body fluids (hemolymph (H), pallial fluid (PF)) are handled in a different way since microcollection tubes have a limited capacity of 1.2 ml. Centrifuge at 750 g for 5 min at 4°C. On ice, remove the supernantant and either refreeze the pellets directly in liquid nitrogen followed by storage at -80°C or add immediately lysis buffer RLT containing βME and process to nucleic acid extraction

Nucleic acid extraction

2 Clean working spaces and pipettes with RNase away before starting. Work under a fume hood.

- Remove samples/microcollection tubes from -80°C and keep on ice
- Remove and discard strip caps from collection tubes
- Add 350 µl buffer RLT+ βME to each tube using a multichannel pipette. Make sure that the tissues are well immersed in the lysis buffer. Close collection tubes with new strip caps.
- Proceed to sample disruption in TissueLyser (3 times 1 min at 30 Hz at room temperature (RT)). Change the plate orientation in the TissueLyser in ordert to assure homogenous sample disruption.
- Centrifuge the lysates from the tissue samples “DG and G” for 5 min at 2250g at 22°C.

Note

Note, that the supplier's manual recommends a centrifugation speed of 5600 g, but due to the lack of such a centrifuge, we have used the maximum speed of 2250 g without further complications.

- Take 100 µl of the supernatant and add to new collection microtubes containing 250 µl of buffer RLT + βME. Mix by pipetting up and down a few times

Note

This step is done in order to avoid clogging of the columns which are adapted for maximum 10 mg of tissues.

- Transfer lysates to an AllPrep 96 DNA plate placed on a new S-block according to plate layout, seal the plate with an airpore tape sheet and centrifuge for 10 min at 2250g. The unused lysate of "tissue samples DG and G" from the previous step is stored at -80°C.
- The AllPrep96 DNA plate is placed on a new S block and stored at 4°C for later DNA extraction
- The S block containing the flow-through is kept and 350 µl of 70% EtOH is added to each well of the S block and immediately mixed by pipetting up and down 3 times. Transfer the mix to an RNeasy 96 plate placed on a new S block.
- Transfer samples to the RNeasy 96 plate, seal with an airpore tape sheet and place in a new S block
- Centrifuge at 2250 g for 5min.
- discard the flow-through in the S block, remove the airpore tape sheet and add 800 µl of buffer RW1 and seal with an airpore tape sheet
- Centrifuge at 2250 g for 5 min
- discard the flow-through in the S block, remove the airpore tape sheet and add 800 µl of buffer RPE and seal with an airpore tape sheet
- Centrifuge at 2250 g for 5 min
- discard the flow-through in the S block, remove the airpore tape sheet and add 800 µl of buffer RPE and seal with an airpore tape sheet
- Centrifuge at 2250 g for 15 min
- place the RNeasy plate on top of the elution microtube rack and remove the airpore tape sheet
- add 45 µl of RNase-free water to elute the RNA. Close the plate with an airpore tape sheet and incubate 1-2 min at RT.
- Centrifuge at 2250 g for 5 min
- add 45 µl of RNase-free water to elute the RNA. Close the plate with an airpore tape sheet and incubate 1-2 min at RT.
- Centrifuge at 2250 g for 5 min
- take an aliquot of the RNA for later measurements of concentration, quality, DNase treatment and cDNA synthesis and close the elution microtube rack with strip caps provided with the kit.

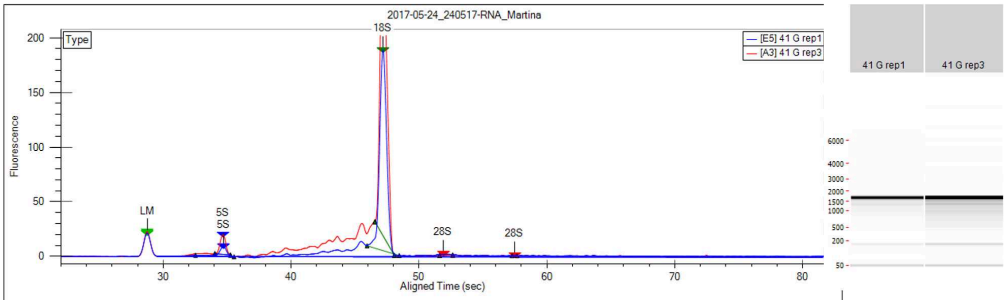
- Store at -80°C.
- For DNA extraction, add 600 µl of buffer AW1 to the AllPrep 96 DNA plate and seal with an airpore tape sheet.
- Centrifuge at 2250 g for 5 min
- discard the flow-through in the S block, remove the airpore tape sheet and add 800 µl of buffer AW2 and seal with an airpore tape sheet
- Centrifuge at 2250 g for 15 min
- place the AllPrep 96 DNA plate on top of the elution microtube rack and remove the airpore tape sheet
- add 50 µl of prewarmed buffer to elute the DNA. Close the plate with an airpore tape sheet and incubate 1-2 min at RT.
- Centrifuge at 2250 g for 5 min
- add 50 µl of prewarmed buffer to elute the DNA. Close the plate with an airpore tape sheet and incubate 1-2 min at RT.
- Centrifuge at 2250 g for 5 min
- take an aliquot of the RNA for later measurements of concentration, quality and close the elution microtube rack with strip caps provided with the kit.
- Store at -80°C.

Quantity/quality assessment of extracted nucleic acids

3 DNA and RNA concentration and purity are measured on 2 µl aliquots using a nanodrop spectrophotometer. The integrity of the RNAs can furthermore be assessed via Bioanalyzer of LabChip.

Note

Note that oyster RNA does not show a 28S RNA peak in the electropherogram. This is most likely due to a thermoconversion of the 28S rRNA during sample preparation (heating) for Bioanalyzer/LabChip analysis and has also been described for insect RNA (doi: **10.1673/031.010.14119**) and snail RNA (**DOI:10.1371/journal.pone.0148390**).



Typical electropherograms (red and blue) and virtual gel image of two *O. edulis* gill RNA samples.